



Exploring effortless AI-generated gamified quizzes in an online special education module: evaluating question quality, student engagement, and its potential to identify at-risk students

Ahmet Durgungoz^{1,2} · Fatma Canan Durgungoz^{3,4}

Received: 18 October 2024 / Accepted: 20 August 2025 / Published online: 19 September 2025
© The Author(s) 2025

Abstract

This study investigates the integration of Artificial Intelligence Generated (AI-generated) gamified quizzes in an online special education module in higher education, focusing on question quality, student engagement, and its potential to identify at-risk students. AI-generated quizzes were developed using Quizizz. These questions were reviewed by experts in special needs education to evaluate their quality in terms of clarity, relevance, cognitive demand, applicability, and feedback potential. Engagement data for different teaching materials, including AI-generated quizzes, instructional documents, and lecture recordings, were collected and analysed over seven weeks to assess their relative effectiveness in maintaining student engagement. Additionally, semi-structured interviews were conducted to gather qualitative insights into student preferences, interactions, and perceptions regarding the various learning tools used in the module. The findings revealed that the majority of AI-generated questions were suitable for direct use, while a notable proportion required modification and a smaller share was excluded. AI-generated gamified quizzes produced an initial significant boost in student engagement, followed by a decline over time, indicating a need for adaptive strategies to sustain interest. Also, it was found that lower quiz performance correlated with poorer exam outcomes, highlighting the questions' predictive potential for identifying at-risk students and enabling timely interventions. Students appreciated the interactive nature and immediate feedback of the AI-generated quizzes, which helped them identify areas for improvement. This study concludes that AI-generated gamified quizzes have significant potential for creating and enhancing student engagement and reducing the workload for instructors. However, human oversight remains crucial to addressing limitations in cognitive demand and the quality of feedback. The findings provide valuable insights into how time-constrained academics can use AI-driven learning tools to create effortless, engaging and effective educational materials.

Keywords AI-generated content · Gamification · Student engagement · Special education · At-risk students · Generative AI

1 Introduction

Integrating Artificial Intelligence Generated Content (AIGC), particularly automated question generation, has emerged as a promising avenue for enhancing teaching and learning. Recent studies have examined the effectiveness of these AI tools, challenges, and potential, revealing a complex landscape that necessitates further exploration (Zhang & Huang, 2024; Durgungoz & Kharrufa, 2025). Generative AI tools like ChatGPT, DeepSeek, Gemini, Claude and Grok have demonstrated significant potential in generating educational content (Ahmed et al., 2024) such as questions, cases, basic digital games, and interactive interfaces that align with learning objectives. For instance, in medical education, AI-generated multiple-choice questions (MCQs) were comparable in quality to those created by experts, though with certain limitations. ChatGPT was noted for generating simpler, more straightforward questions, while Gemini's output was more complex and analytical, indicating that different AI models may suit different educational needs (Ahmed et al., 2024). AI-generated quizzes could, therefore, reinforce knowledge and assess understanding in large, diverse student populations, potentially reducing the workload for educators who are often pressed for time.

However, several studies have highlighted challenges associated with AIGC. Taani and Alabidi (2024) observed that while AI tools could generate questions and explanations that enhance student engagement, they often produced inaccurate or irrelevant content. Similar concerns were raised in medical education contexts, where AI-generated Script Concordance Tests (SCTs) were less nuanced than those created by human experts (Hudon et al., 2024). These findings underline the need for careful review and adaptation of AIGC to ensure its alignment with learning outcomes. Furthermore, using generative AI in education has ethical and practical challenges. Concerns about biases in AIGC¹ and the potential for over-reliance on AI have been widely discussed in the literature (Yan et al., 2023). The systematic review by Ali et al. (2024) emphasised that while AI tools like ChatGPT can generate content quickly and at scale, the risk of introducing biased or ethically questionable material into educational resources remains a significant challenge. This concern is further mentioned by the findings of Laohawetwanit et al. (2024), who noted that although AI-generated tests were generally accurate, they often required substantial refinement to meet the cognitive and ethical standards.

Despite these limitations, there are still advantages to using AIGC. The ability of AI tools to provide immediate, personalised feedback and to adapt content based on individual needs has been highlighted (Meşe et al., 2024). This capability enhances the learning experience by providing tailored support and allows educators to focus on more complex pedagogical tasks. The potential for AIGC to supplement traditional resources, for instance, in fields like radiology, where it can provide dynamic

¹ <https://restofworld.org/2023/ai-image-stereotypes/>.

and interactive learning experiences, was found promising (Meşe et al., 2024). Moreover, studies have shown that AIGC can be particularly effective in environments with low learner engagement or limited resources. Denny et al. (2023) demonstrated that students perceived AI-generated learning resources as similar to those created by their peers. Zhang and Huang (2024) also emphasise the benefits of adaptive gamified assessments, which can tailor content to the learner's proficiency. Such personalisation is particularly relevant in extensive online courses where the diversity of learners and the scale of required content can be challenging for instructors to manage manually.

The potential of AI to create adaptive, personalised learning experiences that cater to individual learner profiles is well-documented (Sekli et al., 2024), and there are initiatives by LLM providers to facilitate students' learning, such as ChatGPT's study mode², which aids learners in understanding a subject progressively, rather than simply providing a straightforward answer. However, while previous research has primarily focused on general AIGC, there is limited empirical evidence addressing the use of AI-supported gamified quizzes, specifically in special education and non-English language contexts. This study extends prior work by evaluating the quality of AI-generated questions, assessing their effectiveness in engaging students and predicting academic risk.

2 Engagement issue in online learning: a case of an AI-supported online platform and gamification

Emerging technologies have significantly expanded opportunities for online learning; however, there are still challenges, such as maintaining student engagement and high dropout rates. Numerous studies have highlighted such issues, noting that the absence of in-person interaction can lead to reduced motivation, isolation, and diminished participation (Park & Kim, 2021). Eichelberger and Ngo (2018) stated that student satisfaction in online learning can be influenced by several factors, such as how the course is organised, how students interact with each other and their instructors, and how well they engage with the course materials. Student engagement, a critical factor for effective learning, is often compromised in remote settings due to the lack of direct interaction with instructors and peers, resulting in passive learning behaviours and lower retention rates (Sercemeli & Baydas Onlu, 2023). Also, the number of neurodivergent students in higher education is increasing, and many face distinct learning challenges. Neurodivergence refers to natural variations in how people's brains work, learn, and process information. It includes conditions like Autism Spectrum Disorder (ASD), Attention Deficit Hyperactivity Disorder (ADHD), Specific Learning Disorder (SLD), Developmental Language Disorder (DLD) and others, and such challenges necessitate inclusive and innovative approaches to enhance student involvement and ensure quality learning outcomes in education (Durgungoz & Durgungoz, 2025).

² <https://openai.com/index/chatgpt-study-mode/>.

Recent literature underlines that gamified strategies such as points, badges, and leaderboards promote active participation, enhance motivation, and foster self-regulatory behaviours (Maimaiti & Hew, 2025; Cespón & Toyos, 2025). A longitudinal study by Maimaiti and Hew (2025) demonstrated that integrating gamification into flipped classrooms improved students' academic achievements, including goal setting and time management. Cespón and Toyos (2025) similarly reported high satisfaction and improved motivation among pre-service teachers engaging with narrative-driven, gamified online activities, although they noted that fostering social interaction within gamified environments remains challenging. Additionally, a bibliometric analysis (Mushtaq et al., 2025) revealed increasing research interest in gamification, particularly regarding its integration with adaptive systems, and its application in hybrid and resource-constrained educational environments. However, despite these promising insights, existing literature largely addresses general educational contexts without a direct focus on special education.

Gamification is a promising approach to enhancing student motivation and engagement (Göksün & Gürsoy, 2019). For example, Park and Kim (2021) found that gamified online learning activities increased student motivation and reduced the stress associated with online courses. Quizizz (rebranded as Wayground in June 2025), with its AI-powered enhancements, offers a compelling example of gamification in action. Ngoc (2024) reports that Quizizz's interactive quizzes, competitive mechanics and instant, personalised feedback substantially increase student motivation and engagement, leading to improved learning outcomes. Hernanz et al. (2024) found that gamified tools like Quizizz can increase students' intrinsic motivation, satisfaction, and performance. Similarly, using platforms like Kahoot and Quizizz has transformed routine activities into engaging educational experiences, enhancing student participation and interest (Bicen & Kocakoyun, 2018). Building on these insights, the present study explores how AI-generated questions on the Quizizz (Wayground) platform can assist academics in creating questions in languages other than English, thereby filling a gap in both gamification and special education research.

3 Research context

This study focuses on a programme where students from various faculties who have chosen to pursue a teaching qualification alongside their primary academic discipline participate in an online module on special needs education. The voluntary nature of the programme means that students are not required to complete the module as part of their core degree requirements. However, these students pay a modest fee to enrol in the programme, adding another layer of motivation. While the fee is not substantial enough to create a significant financial motivation, it introduces a sense of investment and commitment. This unique context creates an ideal platform for this research's aims as the context mirrors the dynamics of lifelong learning and continuous professional development, where individuals, often professionals, seek to enhance their skills through flexible, often online, educational opportunities, such as Massive Open Online Courses (MOOCs).

4 Research questions

The following research questions are formulated to guide this study based on the literature review and the gaps identified:

RQ1: Do AI-generated gamified quizzes align with the learning outcomes for special-education content in a non-English language?

RQ2: Does using AI-generated gamified quizzes in an online special education module enhance student engagement?

RQ3: Can AI-generated gamified quizzes in online learning be used to predict students' academic performance and identify those at risk of underperforming?

5 Methodology

5.1 Researchers' positionality

Recognising academics' heavy workloads and resource constraints, we aimed to offer a practical way to enhance online learning. The second author, a special education expert, designed a module to evaluate whether an online program can improve students' knowledge in special education as part of a broader research (Durgungoz & Durgungoz, [in press](#)), adding hands-on insight. The first author, whose research centres on online learning, generative AI, and pedagogy, recognised this context as an opportunity to explore whether an AI-enhanced gamification platform could ease educators' workload while contributing meaningfully to teaching in a specific discipline, such as special education, with minimal effort required from instructors. Together, they assessed an online format replicable to educators with minimal edtech experience, using simple strategies, AI-generated quizzes alongside documents and lecture recordings, to yield replicable guidance for both blended and fully online contexts.

5.2 Demographics

The demographic overview includes 213 students, with a gender split of 77 males and 136 females (Table 1). Students are distributed across different faculties, such as Sciences (85), Sport (52), Tourism (15), Fine Arts (4), Communication (9), Physical Sciences (36), Engineering (3), Nursing (3), and Economics and Administrative Sciences (6).

Participants were undergraduates from non-specialist degree programmes who were taking a module in special education. It is important to note that findings should therefore be interpreted for non-specialist cohorts and not generalised to specialist special-education programmes or professional training settings.

5.3 Procedure

The first step involved generating quiz questions in Turkish using an AI-supported gamified quiz platform, Quizizz. AI-generated MCQs with AI-generated feedback for

Table 1 Demographics of participants

Items	<i>N</i>	%
Total Participants	213 students	100%
Gender Distribution		
- Male (M)	77	36.15%
- Female (F)	136	63.85%
Faculty Distribution		
- Sciences	85	39.91%
- Sport	52	24.41%
- Tourism	15	7.04%
- Fine Arts	4	1.88%
- Communication	9	4.23%
- Physical Sciences	36	16.90%
- Engineering	3	1.41%
- Nursing	3	1.41%
- Economics and Administrative Sciences	6	2.82%
Interviewed Participants Gender Distribution		
- Female	5	71.43%
- Male	2	28.57%
Interviewed Faculty Distribution		
- Sciences	2 (1 M, 1 F)	28.57%
- Sport	2 (1 M, 1 F)	28.57%
- Physical Sciences	1 F	14.29%
- Communication	1 F	14.29%
- Nursing	1 F	14.29%

each question were exclusively requested from the AI tool for consistency in format. The authors utilised a premium membership for the platform, costing approximately \$36 annually, to access its full features. The AI-generated 268 MCQ questions and feedback were created to align with the learning outcomes of the special education module using the module's teaching materials. Two academics (the second author and an academic from a special education department), specialising in special education, reviewed these questions using a rubric. To ensure transparency in data collection and analysis, evaluators underwent discussion sessions prior to scoring. Evaluators independently scored this pilot set, then convened to discuss discrepancies and refine their scoring guidelines. After calibration, they scored the questions independently, and inter-rater reliability was calculated using a percentage agreement, which yielded an agreement rate of 86%.

After the evaluation, the engagement levels with different teaching materials were analysed over seven weeks. The materials included AI-generated gamified quizzes, instructional documents (such as PowerPoint presentations and PDFs), and a video of the instructor discussing the course slides. The number of users engaging with each material type was visualised weekly to compare engagement levels and identify patterns.

5.4 Rubric

The rubric was created by scanning the literature and was subsequently reviewed by four academics, two from Special Education (including the second author), one from Educational Technology (first author), and one from the Education Sciences department, to evaluate criteria such as relevance, clarity, cognitive demand, applicability, variety, innovation, and feedback potential. The evaluators from the Special Education department, with experience preparing undergraduate-level questions, independently assessed the AI-generated questions. The academic from the Education Sciences department, chosen based on his research profile in assessment and evaluation, volunteered to contribute to refining the rubric.

The scoring was adjusted to align with engaging students rather than satisfying all the criteria in the rubric, recognising the limitations of MCQ questions (Table 2). Since the assessment relied on MCQs, we anticipated restricted higher-order demand and accordingly prioritised relevance to learning outcomes, clarity and precision, cognitive demand in the weighting, using a conservative threshold for applicability, variety and innovation, and feedback potential. Our primary aim was to encourage students to interact with the quiz questions, to foster a more comprehensive engagement with the module content. Therefore, Relevance to Learning Outcomes (25 points), Clarity and Precision (20 points), and Cognitive Demand (20 points) were assigned higher maximum points because they are deemed most critical for ensuring that questions are well-aligned with learning outcomes and sufficiently challenging.

In contrast, Applicability (15 points), Variety and Innovation (10 points), and Feedback Potential (10 points) received lower maximum scores, reflecting the module's primary aim of maintaining student engagement rather than promoting advanced

Table 2 Rubric for evaluating AI-generated questions with criteria descriptions and maximum points

Criterion	Description	Points (Max)
Relevance to Learning Outcomes (Biggs, 1996)	The degree to which the question aligns with the specific learning outcomes of the module	25
Clarity and Precision (Stiggins, 1994)	Clarity of language and precision in assessing the intended concept	20
Cognitive Demand (Anderson & Krathwohl, 2001)	Level of cognitive engagement required, based on Bloom's taxonomy	20
Applicability (Merrill, 2002)	Practical application of the question in reinforcing course concepts	15
Variety and Innovation (Torrance & Pryor, 1998)	Diversity of question types and use of innovative approaches to engage students	10
Feedback Potential (Nicol & Macfarlane-Dick, 2006)	Ability of the question to provide meaningful feedback that guides further learning	10

challenge. The literature also evidently shows that AIGC still has limitations (Lao-hawetwanit et al., 2024). The passing score was set at 70 points based on the calibration exercises to ensure that only questions meeting a robust quality threshold were included in the module. Future studies utilising various AI-generated question types can assign equal scores to each criterion in the rubric.

5.5 Data collection

To complement the quantitative data, semi-structured interviews were also conducted to gain deeper insights into student experiences. These interviews combined open-ended surveys, telephone conversations, and email exchanges. The goal was to understand the factors influencing student engagement and preferences, particularly regarding AI-generated gamified quizzes. Interview questions revolved around the benefits and drawbacks of the quizzes, the immediacy and usefulness of feedback, and how these factors influenced overall engagement. Some of the key interview questions were “How did the quizzes support or hinder your learning?” and “In what ways did the feedback influence your motivation or study habits?”. All students who were in the module were invited via email to participate voluntarily in the study. Due to the online nature of the module, only seven students agreed to be interviewed.

As the final step of the analysis, the study examined the relationship between students’ performance on AI-generated gamified quizzes and their final exam results. Students were informed that if they used their real names on the Quizizz platform, their quiz performance data would be compared with their exam scores for research purposes, ensuring voluntary participation and data being anonymised for analysis. As a result, 58 students’ Quizizz performances were documented for analysis.

5.6 Analysis

The quantitative data analysis involved visualising engagement levels with different teaching materials over time, providing a comparative view of how each type of material was utilised throughout the seven weeks within an eight-week teaching period. This included running statistical analyses on user engagement trends and click rates for quizzes, documents, and videos to assess their relative effectiveness in maintaining student interest. These data were derived from the Virtual Learning Environment for analysis. To assess the quality of AI-generated questions, the scores from the rubric-based evaluation were analysed to identify strengths and weaknesses in the AI-generated questions and feedback.

Concept mapping was used to analyse semi-structured interviews (Striepe, 2021). Incorporating concept mapping in the analysis improves validity by offering a more trustworthy representation of the participant’s perspective (Scherp, 2013). This thematic analysis helped contextualise the quantitative findings and provided a richer understanding of how students interacted with AI-generated quizzes and other course materials.

Finally, the relationship between quiz performance and exam results was analysed to explore the predictive value of AI-generated quizzes in identifying at-risk students. This analysis aimed to determine whether students’ performance on gamified quizzes

could serve as an early indicator of their likelihood to struggle in the exam, allowing for timely interventions and support. We analysed exam scores using Python, leveraging the seaborn and matplotlib libraries to create a violin plot that visualises the distribution of scores between two groups: “High Score Quizizz” and “Low Score Quizizz”. This plot was selected because it effectively combines box and density plots’ features, offering a comprehensive view of the data distribution. Additionally, the smoothing technique used in the violin plot can suggest a probability of scores in a range where no actual data points exist. For example, in our analysis, the smoothing extended the density curve up to 75 in the Low Score Quizizz group, although no students achieved that score. This highlights potential areas where scores could cluster, offering a better understanding of the distribution, even if those exact scores are not present in the dataset.

6 Results

6.1 Evaluation of AI-generated gamified quiz questions

A total of 268 MCQs were generated using AI, each paired with auto-generated feedback to support students’ understanding of key topics in Special Education. The questions were independently reviewed by two expert evaluators and classified into three categories: (1) directly used without modification, (2) modified for suitability, and (3) excluded. Figure 1 presents the rounded percentage distribution of these categories across each topic area. Each stacked bar sums to 100%, and the percentage labels printed within the segments show the share of questions that were used directly, modified for suitability, or excluded. 12% questions were excluded due to reasons such as duplication, poor phrasing, or lack of relevance to the course outcomes. 31%

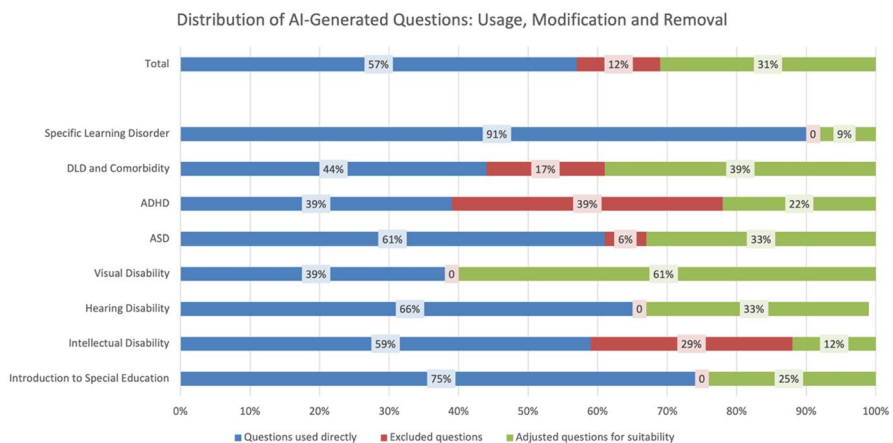


Fig. 1 Distribution of AI-generated MCQs across Special Education topics. Each horizontal bar represents the percentage of questions categorised as directly used (blue), modified for suitability (green), or excluded (red). Percentages are calculated per topic area, with each bar totalling 100%

were modified to improve clarity or Alignment with course content. The remaining 57% were used without any changes. These results suggest that AI-generated gamified quizzes can generate a significant proportion of questions. However, excluding 12% questions and modifications required for 31% also indicates that while the AI tool is effective, the algorithms/prompts can be improved to ensure the accuracy of questions and greater alignment with specific learning outcomes.

The two experts further assessed the set of questions generated per topic (comprising both directly included and modified questions) against the rubric. Figure 2 shows the average scores of AI-generated quiz questions per topic across six evaluation criteria. AI-generated quiz questions scored highest in Relevance to Learning Outcomes ($M=21.25$, $SD=1.06$) and Clarity and Precision ($M=18.00$, $SD=1.13$). Lower scores were observed in Cognitive Demand ($M=6.00$, $SD=0.71$), Applicability ($M=8.50$, $SD=0.85$), Variety and Innovation ($M=5.50$, $SD=0.57$), and Feedback Potential ($M=5.00$, $SD=0.57$). The minor standard deviations across all criteria indicate strong agreement between the two expert raters, as seen in Fig. 2. These patterns are consistent with the inherent constraints of MCQs, which prioritise recognition and recall rather than analysis, application or evaluation in Bloom's taxonomy.

The low score in "Cognitive Demand" reveals that the questions were not designed to challenge students' higher-order thinking skills but rather focused on information retrieval. In our context, this reflects format rather than only content quality, since an MCQ-only design limits opportunities for higher-order reasoning and extended justification. Similarly, the scores for "Variety and Innovation" and "Feedback Potential" indicate that the AI-generated questions lacked diversity and were limited in providing constructive feedback that could guide further learning. These limitations

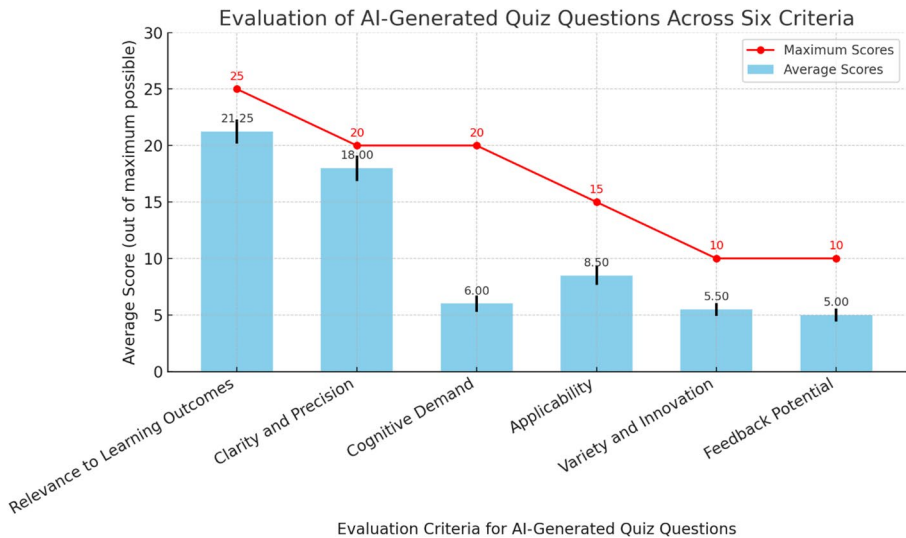


Fig. 2 Average expert ratings of AI-generated quiz questions across six evaluation criteria. Blue bars represent mean scores, while red dots indicate the maximum possible scores. Error bars show standard deviations based on ratings from two expert reviewers, reflecting the level of agreement between them

emphasise the need for a balanced approach considering cognitive challenge and feedback quality.

In addition to evaluating the quality of the AI-generated questions, we also measured the time efficiency of creating these quizzes. Over a period of seven weeks, seven quizzes were prepared, with the academic spending an average of 20–25 min per quiz. This time included uploading the relevant documents to the AI platform, generating initial questions, refining the language, adding supplementary AI-generated questions where necessary, and making final edits. The relatively low time investment required for each set of quizzes suggests that AI-supported platforms can significantly reduce the workload for educators in creating engaging and educationally aligned assessments.

6.2 Student engagement: AI-generated quiz vs. other teaching materials

In this section, students' engagement with teaching materials over a seven-week period will be presented, providing insights into the usage patterns of different teaching materials. The first week was not included in the analysis due to late enrollments. Figure 3 illustrates how many users engage with three distinct teaching materials each week: documents, videos (lecture recordings), and gamified quizzes. To evaluate whether student engagement differed significantly by teaching material type, a one-way ANOVA was conducted on the number of users interacting with documents, videos, and gamified quizzes over seven weeks. The analysis revealed a statistically significant difference in engagement levels, $F(2, 18)=66.44$, $p<.000$. This suggests that the type of material had an influence on student engagement, with gamified quizzes consistently attracting higher user interaction compared to documents and videos.

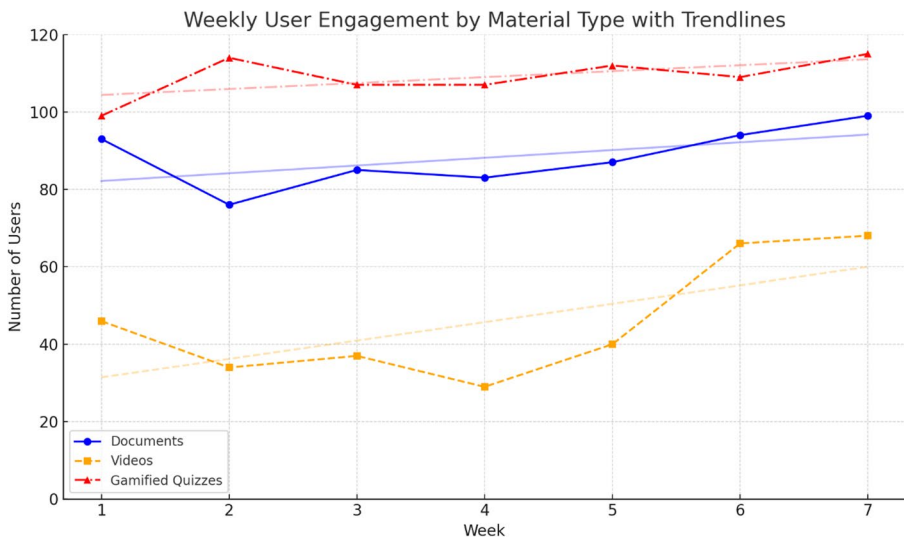


Fig. 3 Line chart showing the number of engaged users for different types of teaching materials over seven weeks, showcasing documents (blue circles, solid line), lecture recordings (yellow squares, dashed line), and AI-generated questions (red triangles, dotted and dashed line)

Figure 3 shows that the AI-generated gamified quiz consistently attracted the highest number of users across all weeks, starting with over 100 users in Week 1 and maintaining a steady level of engagement with a slight upward trend by Week 7. To examine the patterns of user engagement over time, a linear regression analysis was conducted for each type of teaching material across seven weeks. The results indicated a significant decline in individual user engagement with documents ($\beta = -4.93$, $p = .036$, $R^2 = 0.60$), reflecting a steady decrease in students accessing document-based materials. For videos, the regression also showed a negative slope ($\beta = -2.21$), though it was not statistically significant ($p = .125$, $R^2 = 0.39$). Gamified quizzes demonstrated a slight decline in user numbers ($\beta = -3.71$), which was likewise not statistically significant ($p = .067$, $R^2 = 0.49$).

Figure 4 shows the total number of clicks recorded for each material type over the same period. This provides a different perspective by highlighting weekly clicks on each material type, reflecting overall user engagement and interaction intensity.

To examine engagement patterns, a one-way ANOVA was conducted on the total number of clicks for documents, videos, and gamified quizzes over seven weeks. The results revealed a significant difference between material types, $F(2, 18) = 10.79$, $p = .001$. This suggests that students clicked on gamified quizzes substantially more than videos and documents, indicating greater engagement with interactive formats.

To further investigate the observed decline in clicks over time, a linear regression analysis was conducted for each type of teaching material across seven weeks. The analysis revealed a significant decrease in clicks for documents ($\beta = -40.82$, $p = .045$, $R^2 = 0.59$). However, the decline may reflect reuse of previously downloaded materials, which may have reduced the need for repeated access. Videos also showed a negative trend ($\beta = -13.57$), though this was not statistically significant ($p = .172$, $R^2 =$

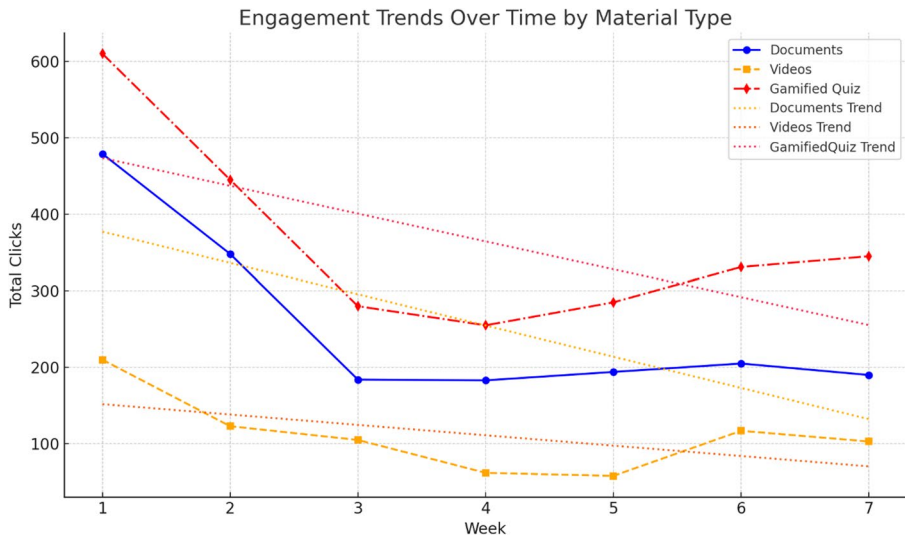


Fig. 4 Line chart showing total clicks on different teaching materials across seven weeks showing documents (blue circles, solid line), Lecture recordings (yellow squares, dashed line), and AI-generated questions (red diamonds, dotted and dashed line)

0.34). For gamified quizzes, a visible decline was noted ($\beta = -36.36$), but again, this was not statistically significant ($p = .131$, $R^2 = 0.40$).

6.3 Correlation Between AI-generated Quiz and Exam Performance: A Comparative Analysis of High and Low Scoring Students

In this section, we examine whether the high levels of engagement observed with the gamified quizzes translated into meaningful outcomes. As will be discussed in the next section, the interviews suggested that these interactive elements effectively captured student interest and facilitated engagement with other materials; however, it remains unclear whether this engagement provided instructors with insights into designing a better environment for online learning. To explore this, we compared the exam results of students who scored high on the quizzes with those who scored low to determine if frequent interaction with the gamified quizzes correlates with improved academic performance. Students were classified into High and Low Score Quizizz groups using the faculty's official pass mark of 60 as the cut-off, in line with institutional grading policy (CC or above). This threshold was selected to ensure consistency with established academic standards. We sought to investigate whether there is a predictive relationship between students' performance on the gamified quizzes and their subsequent exam results. The results revealed a clear distinction between the two groups, as presented in Fig. 5.

A Pearson correlation analysis was conducted to examine the relationship between students' Quizizz scores and their exam performance. The results showed a strong

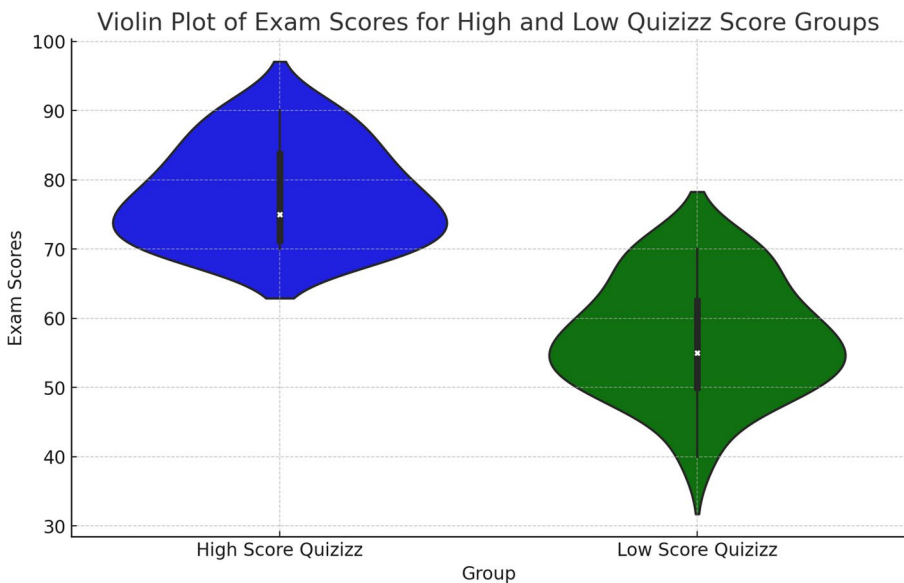


Fig. 5 Violin plot comparing exam score distributions between two quizizz score groups, classified by a median split on each student's average quizizz percentage across all quizzes: high score group (blue) and low score group (green). The violins display the density and full range of scores, with white dots marking the median and black bars indicating the interquartile range

and statistically significant positive correlation ($r=.68$, $p<.001$), indicating that students who performed well on Quizizz generally achieved higher scores in the final exam. Students in the High Score Quizizz group had exam scores between 70 and 90, with the majority scoring between 70 and 80. This suggests that students who engaged more effectively with the Quizizz platform performed better on their exams. The consistency in their exam scores indicates that high-quality engagement with the gamified learning tool predicts academic success. The Low Score Quizizz group displayed a more comprehensive range of exam scores, from 40 to 70, with most students clustering between 50 and 60. While there is no evidence that engagement with AI-generated questions leads to better outcomes, such a correlation is still valuable, as scoring low in such AI-generated quizzes can serve as a valuable predictor of student performance, allowing instructors to first engage at-risk students, identify them, and take precautions.

6.4 Students' perspective of self-directed learning and gamified quizzes: preference for flexible, interactive and fun learning except for long lecture recordings

Three key themes emerged from the in-depth interviews, as shown in Fig. 6. To create this concept map, we first conducted a thematic analysis of the interview transcripts, identifying recurring ideas and grouping them into higher-order categories. The colour-coded bigger nodes (blue, green, and orange) represent these main themes; (1) Flexibility and Accessibility, (2) Need for Interaction and Dynamic Content, and

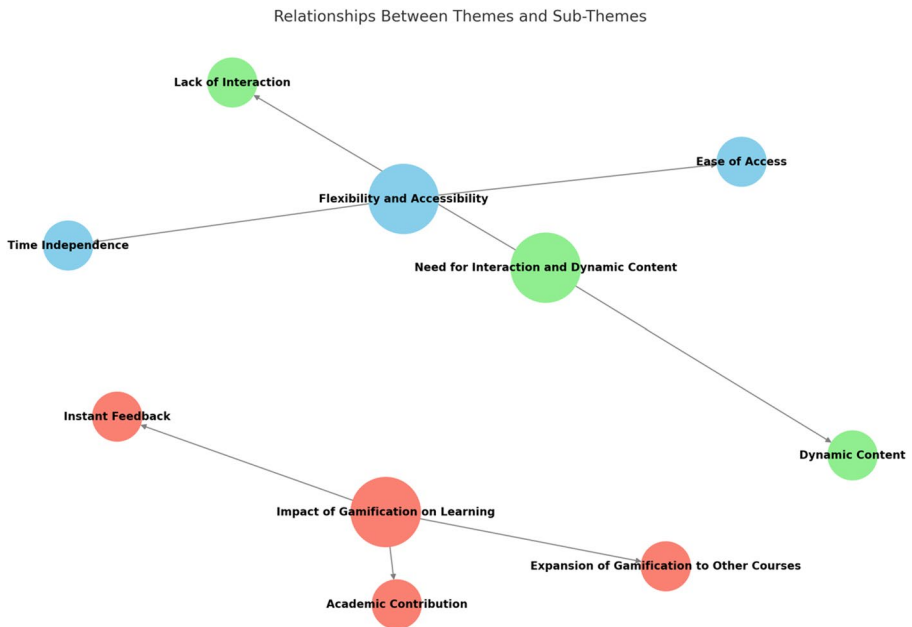


Fig. 6 Concept map illustrating the relationships between main themes and their sub-themes

(3) Impact of Gamification on Learning, while the smaller nodes indicate sub-themes that emerged during coding. The arrows depict the direction and strength of relationships between themes, showing how sub-themes contribute to broader thematic concepts.

6.5 Flexibility and accessibility

Participants stressed flexibility as online learning's key benefit, especially for those with hectic timetables. *"The biggest advantage was having access to course materials at any time"* (P2), and *"I could access the course materials whenever I wanted and did not have to go to the faculty of education for classes"* (P5). Ease of access also aided students juggling multiple modules and work: *"Remote learning was the best side; this was the best thing for us because we have to take many modules here and there"* (P4), while one working student noted, *"I study and work at the same time, so having this module online was very good"* (P6). By contrast, long lecture recordings proved hard to fit in: *"I just couldn't have an hour to sit down and watch a whole lecture when I'm trying to cram for my other classes"* (P4). Many recommended shorter, 5–10 min segments *"I could watch [...] on the bus or between classes"* (P5).

6.6 Need for interaction and dynamic content

Participants emphasised Quizizz's role in making learning more interactive and motivating. *"Those questions made learning fun and somehow allowed me to interact with you as I knew you were checking the leaderboard. It kept me motivated throughout the course"* (P2). P3 added that quizzes helped highlight tricky bits for later review: *"Quizizz especially helped me understand tricky bits through questions...those questions helped when I later read the slides"* (P3), while P5 appreciated the immediate feedback: *"I was able to see where I needed to improve and understand the correct answers immediately"* (P5).

They also found Quizizz more engaging than traditional materials. *"I would definitely like to see tools like Quizizz used in other courses... it was like reading a course document"* (P4). Even when the topic lay outside their main interest, one student confessed, *"I wished the questions never ended...and we need to learn these things to be a better teacher so I can say that webpage [Quizizz] was helpful"* (P1). Overall, gamified quizzes boosted interactivity, aided comprehension and provided timely feedback.

6.7 Impact of gamification on learning

Participants noted that such gamified activities make learning more enjoyable and help them better understand the subjects. For instance, P4 stated that *"Quizizz made me go and look for more information from the documents and YouTube, especially helped me understand certain bits"* (P4), expressing the benefit of such applications. The instant AI-generated feedback helped students recognise their mistakes. As P5 stated, *"The feedback after each question was like an extra reading..."*. Another student noted, *"It was like questions that sometimes influencers put on Instagram, so yes, that was different in a good way"* (P6). The interviews highlighted the significant

role that gamified quizzes play in capturing student attention and fostering engagement with the teaching materials, as noted by P1:

Another teacher of mine used a similar thing just once. It was like a game; it tricks you and covers the boring bits, so no matter what the topic is, you want to be on the leaderboard.

P7 student remarked, “*The quizzes were fun and made me curious, so I ended up reading the documents and watching some videos afterwards.*” Such self-reports suggest that gamified content can act as a facilitator, motivating students to explore and engage with other resources.

7 Discussion

The findings from this study offer valuable insights into the effectiveness of AI-generated gamified quizzes in enhancing student engagement and identifying at-risk students. While the quizzes demonstrated the ability to boost initial engagement and provide a more interactive learning experience, several key challenges and limitations warrant further exploration. The study’s implications for predicting at-risk students, maintaining long-term engagement and ensuring content quality will be discussed. The potential of AI in supporting educators, particularly regarding workload reduction, will also be discussed, alongside the need for continuous human oversight to address content and cognitive challenges.

7.1 Effectiveness and limitations of AI-generated gamified quizzes

The findings of this study highlight both the potential and limitations of using AI-generated gamified quizzes to Align with learning outcomes in special education. Our results show that 57% of the AI-generated questions met the required standards for direct inclusion and 31% after modification, demonstrating the tool’s capacity to produce relevant questions and a starting point for instructors who struggle to generate questions. The highest scores were achieved in “Relevance to Learning Outcomes” and “Clarity and Precision,” suggesting that AI tools can effectively align questions for clear information retrieval. This aligns with previous studies that found AIGC well-aligned, such as in dental education (Ahmed et al., 2024). However, it should be noted that the participant students were not from the School of Education, so the content and the questions generated aimed to deliver the basics of Special Education.

An important strength of AI-generated material is its scalability. AI-generated quizzes are inherently scalable because they are browser-based and can be accessed on any device. This minimal reliance on technical infrastructure makes such tools particularly suitable for under-resourced settings where financial and technical resources may be limited. Moreover, the ease of access and low setup requirements suggest that the approach can be readily generalised to other educational contexts and disciplines beyond special education, offering a cost-effective method to enhance student engagement and adaptive assessment across diverse learning environments.

It should be noted that certain limitations were also evident in the AIGC, particularly in the lower scores for “Cognitive Demand,” “Applicability,” “Variety and Innovation,” and “Feedback Potential.” These findings indicate that while the AI-generated quizzes successfully created straightforward, engaging questions to maintain student interest, they fell short in fostering cognitive engagement. This is consistent with research suggesting that AI tools may not fully capture the complexity required for higher-order thinking skills (Taani & Alabidi, 2024). Hernanz et al. (2024) noted that while gamified tools like Quizizz increased engagement, their effectiveness varied across different disciplines. Specifically, students in more practically oriented fields benefited more from such tools, highlighting the importance of adapting AIGC to the specific cognitive demands of different subjects. Tandiono (2024) pointed out that while tools like Kahoot enhanced student engagement, they often lacked the depth needed for meaningful interactions. This reinforces the need for expert oversight in using AIGC to ensure that AI-generated quizzes foster deep learning rather than just immediate engagement. The limited variety and feedback capacity also reflects findings from studies in medical education, where AI-generated tests required significant refinement to meet educational standards (Laohawetwanit et al., 2024). Zhang and Huang (2024) underlined that adaptive gamified tests significantly reduced learner dissatisfaction and enhanced learning outcomes, suggesting that AI-generated quizzes could be more effective if they included adaptive features. This aligns with our findings regarding the limitations in “Variety and Innovation,” highlighting the potential benefits of AI tools that adjust to individual student needs to provide a more tailored learning experience.

Another important practical benefit was the time efficiency of creating quizzes using AI-powered tools. Our results showed that the academic could prepare a quiz for around 20–25 min, highlighting the tool’s capacity to alleviate educators’ workload. This is particularly valuable in designing online or in-class activities, where academics must manage multiple responsibilities. The ability to rapidly generate quiz questions mostly aligned with learning outcomes could offer a sustainable way to integrate more frequent assessments and interventions. Similar to our findings, Hutt and Hieb (2024) also reported how generative AI, specifically through the use of large language models like ChatGPT, can support mastery learning, a pedagogical approach that requires students to achieve a high level of proficiency before moving on to more complex topics. However, the authors also highlight limitations, including the need for high-quality input material and the potential biases in AIGC.

While the findings highlight the potential of AI-generated gamified quizzes to enhance engagement and reduce instructor workload, it is important to consider the ethical limitations of such tools critically. AIGC, including the questions created via Quizizz, is susceptible to bias and inaccuracy stemming from the training data or prompt structure. In our evaluation, certain questions demonstrated ambiguous phrasing or limited cognitive challenge issues that were particularly problematic in the context of special education. For instance, one question included the statement “*Students with learning disabilities cannot benefit from group activities*” as an incorrect option in a multiple-choice format. Although the correct option was indeed accurate, the inclusion of this deficit-oriented statement as a distractor risked reinforcing a harmful misconception. In this case, the problem lay not in the correct answer

but in the binary structure of the question. A more appropriate format would have been a multiple-answer item that allowed for a nuanced evaluation of pedagogical strategies rather than forcing a single-choice response that inadvertently legitimised a problematic viewpoint. Such issues emphasise the necessity of human oversight not only for pedagogical alignment but also for ethical assurance. Without educator intervention, AIGC risks reinforcing stereotypes, excluding marginalised learners, or presenting inaccurate or misleading information. As generative AI becomes more integrated into educational contexts, particularly those involving diverse learner needs, future development must prioritise inclusive design principles and transparent review mechanisms. This is especially crucial for fields like special education, where nuance, sensitivity, and contextual relevance are essential.

7.2 Engagement and student interaction with teaching materials

The engagement analysis revealed that AI-generated gamified quizzes consistently attracted the highest number of users, particularly in the early weeks. The interview results and VLE logs support the notion that gamified quizzes can enhance student engagement by making learning more interactive and enjoyable, which has also been echoed in the literature on using gamified tools in online learning (Sekli et al., 2024). As highlighted in the interviews, the interactive and dynamic nature of the quizzes provided instant feedback, which kept students actively involved. This suggests that the gamified aspect engages students initially and serves as an entry point to other educational resources, promoting a broader engagement with course content.

However, the decline in quiz engagement over time, as shown by the decrease in total clicks, indicates a potential novelty effect that may wear off, reducing sustained engagement. The observed decline in engagement after Week 4 aligns with patterns often seen in MOOC-like learning environments, where initial interest tends to taper off over time. Given that students enrolled in this module as a means of career diversification rather than a core requirement, a gradual drop in participation was anticipated. This suggests that maintaining consistent engagement in optional, online modules may require additional strategies, such as dynamically integrating gamified elements, instructor presence or personalising feedback to sustain motivation. Interviews also showed that the gamified format drew students in at first, then interest settled. To keep participation up once the novelty wears off, shorter, tightly focused videos and adaptive quiz sets with fewer but more targeted items can be used.

The initial high engagement observed in our study is consistent with findings from Abbas et al. (2024), who noted that metrics like the number of questions answered and correct responses strongly correlated with early engagement. However, the subsequent decline in interaction may reflect what Abbas et al. (2024) described as a novelty effect that wears off over time, suggesting the need to continuously adapt the learning materials to sustain engagement. This decline aligns with findings from studies indicating that online learning without instructor presence or other motivation factors could lead to high dropout rates (Richardson et al., 2015). Furthermore, Keadkraichaiwat et al. (2024) highlighted the effectiveness of team-based interactive activities, like Kahoot-T, in maintaining enthusiasm and peer competition. This suggests that incorporating more team-based or collaborative elements into AI-generated

quizzes might help sustain engagement beyond the initial novelty phase by catering to diverse student preferences. The low engagement with lecture recordings, particularly due to their length, further stresses the need for more dynamic and flexible content delivery methods. This highlights a potential role for AI video generator platforms to split longer lecture recordings into shorter, more engaging bite-sized parts with integrated questions to maintain student interest and provide continuous feedback loops (Yan et al., 2023).

7.3 Predicting at-risk students through AI-generated gamified quizzes

Our study revealed that the AI-generated quizzes effectively captured students' attention across different performance levels. However, while engagement is a positive outcome, it does not necessarily equate to quality learning. Our analysis revealed that although some students actively engaged with the gamified quizzes, this did not always translate into high academic performance. Specifically, students who interacted with the quizzes but consistently scored low on these assessments also tended to perform poorly on their final exams. This suggests that while AI-generated gamified quizzes effectively draw students in, they may not foster deep understanding or retention of course material for all learners. As also reported in our study and the literature, while gamification can capture student attention and encourage more frequent interaction with learning materials, it may not lead to meaningful learning gains (Balci et al., 2022; Sercemeli & Baydas Onlu, 2023; Cigdem et al., 2024).

Despite their unclear impact on academic performance, AI-generated questions have the potential to predict at-risk students. For instance, Bolat and Taş (2023) found that gamified assessment tools significantly impacted student performance and could serve as valuable predictors of academic outcomes. Our study also identified a correlation between consistently low quiz scores and poor/average exam performance, indicating that AI-generated quizzes could act as an early warning system for at-risk students. This predictive capability is further supported by Zhang and Crawford (2024), who showed that students with higher motivation tended to perform better in gamified environments, suggesting that low quiz scores might signal the need for additional support. The Quizizz's built-in features, including easy-to-access detailed quiz statistics and interaction logs, allow educators to monitor student performance with minimal effort. By examining these data points, instructors can identify engaged yet underperforming students, enabling targeted interventions to address gaps in understanding (Zhang & Crawford, 2024). For example, Zainuddin (2023) highlighted using gamified quizzes to identify students who may struggle with course content, allowing instructors to intervene early and provide additional support. Zainuddin (2023) showed that gamified quizzes can flag students who struggle early on, enabling timely intervention, while Taani and Alabidi (2024) observed that AI-generated quizzes often expose knowledge gaps when performance is poor, indicating where extra support is needed. Hudon et al. (2024) likewise found AI-generated Script Concordance Tests useful for diagnosing misconceptions in medical education. Our findings echo these benefits, AI-generated quizzes are quick to create yet accurately predict and engage at-risk students, allowing educators to intervene promptly with tailored support.

8 Limitations

The study has several limitations. First, the exclusive use of MCQs generated in Turkish may not fully capture students' understanding, with AI tools potentially yielding higher-quality questions in English. Second, MCQs are efficient for large cohorts and align well with rapid feedback, yet they typically elicit recognition rather than explanation. Third, despite careful efforts to ensure content quality by using detailed prompts, the utilised original lesson materials may have influenced the quality of the AI-generated questions. Fourth, it is important to interpret these findings within context, as the participants were not from a School of Education, and the content was intentionally designed to introduce foundational concepts in Special Education. In a more advanced educational setting, where greater cognitive challenge is expected, AI-generated questions' perceived appropriateness and effectiveness may be lower, which could require intensive human insight. Lastly, due to the online format of the module, it was challenging to reach a larger number of students, resulting in a limited pool of interview participants.

9 Conclusion and recommendations

The findings from this study have broader implications, potentially shedding light on behaviours and learning patterns applicable to a more general population engaged in lifelong learning. This study highlights the potential benefits of AI-generated gamified quizzing tools in enhancing student engagement, with implications extending beyond engagement and predicting students at risk of academic failure. Using AI-generated questions to identify academically at-risk students offers a proactive approach, allowing for timely and targeted interventions. Future research can also pair MCQs with AI-generated open-ended or scenario-based items. For instance, short-answer justifications, multiple-choice with an explanation of the learner's choice, or vignette-based decisions would provide more profound insight. Using open-ended, scenario-based questions could be beneficial; however, reviewing their suitability and assessing students' responses would create additional workload in this context.

By prioritising a practical, low-effort approach to online module design, the study offers valuable insights for academics seeking to enhance their teaching without needing extensive budget, expertise and time. However, the limitations identified in our study, particularly regarding cognitive demand, variety, applicability and feedback potential, highlight areas for improvement to maximise the potential of these tools. It is also important to note that the module's voluntary status and modest enrolment fee may have increased baseline motivation through self-selection and commitment effects. Future studies may focus on collecting fee-related variables to examine whether financial investment correlates with engagement.

Additionally, leveraging AI to create varied content types, such as integrating quizzes within shorter, more targeted video segments, could enhance engagement further, addressing the need for more dynamic and flexible learning resources. To address the observed decline in quiz use, future research should explore adaptive AIGC and test strategies for integrating these tools within a broader, more diverse set of teaching

strategies beyond Turkish/English. Furthermore, institutions should consider establishing comprehensive training programs and support systems to facilitate the effective adoption of AI tools, ensuring broader applicability across educational contexts.

Authors' contributions Both authors contributed equally to the design, data collection, analysis and write-up process.

Funding No funding.

Data availability The datasets used and/or analysed are available from the corresponding author upon reasonable request.

Declarations

Ethics approval For this study was obtained from Mersin University[REF: 29]. Participation was voluntary, and anonymity was ensured throughout the data collection and analysis.

Competing interests No competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Abbas, R., Sebastián, A., & Casanova, J. (2024). Evaluating classroom response systems in engineering education: Which metrics better reflect student performance? *Education and Information Technologies*, 1–9. <https://doi.org/10.1007/s10639-024-13048-5>
- Ahmed, W. M., Azhari, A. A., Alfaraj, A., Alhamadani, A., Zhang, M., & Lu, C. T. (2024). The quality of AI-Generated dental caries multiple choice questions: A comparative analysis of ChatGPT and Google bard Language models. *Heliyon*, 10(7), 1–9. <https://doi.org/10.1016/j.heliyon.2024.e28198>
- Ali, D., Fatemi, Y., Boskabadi, E., Nikfar, M., Ugwuoke, J., & Ali, H. (2024). ChatGPT in teaching and learning: A systematic review. *Education Sciences*, 14(6), 643. <https://doi.org/10.3390/educsci14060643>
- Anderson, L. W., & Krathwohl, D. R. (Eds.). (2001). *A taxonomy for learning, teaching, and assessing: A revision of bloom's taxonomy of educational objectives*. Longman.
- Balci, S., Secaur, J. M., & Morris, B. J. (2022). Comparing the effectiveness of badges and leaderboards on academic performance and motivation of students in fully versus partially gamified online physics classes. *Education And Information Technologies*, 27(6), 8669–8704. <https://doi.org/10.1007/s10639-022-10983-z>
- Bicen, H., & Kocakoyun, S. (2018). Perceptions of students for gamification approach: Kahoot as a case study. *International Journal of Emerging Technologies in Learning*, 13(2), 72–93. <https://doi.org/10.3991/ijet.v13i02.7467>
- Biggs, J. (1996). Enhancing teaching through constructive alignment. *Higher Education*, 32(3), 347–364. <https://doi.org/10.1007/BF00138871>
- Bolat, Y. I., & Taş, N. (2023). A meta-analysis on the effect of gamified-assessment tools on academic achievement in formal educational settings. *Education and Information Technologies*, 28, 5011–5039. <https://doi.org/10.1007/s10639-022-11411-y>

- Cespón, M. T., & Toyos, P. B. (2025). An example of gamification for pre-service teachers in online higher education: Methods, tools, and purpose. *Digital Education Review*, 46, 1–14. <https://doi.org/10.1344/der.2025.46.1-14>
- Cigdem, H., Korkusuz, M. E., & Karaçaltı, C. (2024). Gamified learning: Assessing the influence of leaderboards on online formative quizzes in a computer programming course. *Computer Applications in Engineering Education*, 32(2), e22697. <https://doi.org/10.1002/cae.22697>
- Denny, P., Khosravi, H., Hellas, A., Leinonen, J., & Sarsa, S. (2023). Can we trust AI-generated educational content? Comparative analysis of human and AI-generated learning resources. *ArXiv Preprint arXiv*, 2306, 10509. <https://doi.org/10.48550/arXiv.2306.10509>
- Durgungoz, A., & Durgungoz, F. C. (in press). The advantages and the shortcomings of self-directed online special education needs module: Addressing the theory, practice and personal experience. *Bartın University Journal of Faculty of Education*.
- Durgungoz, F. C., & Durgungoz, A. (2025). Interactive lessons are great, but too much is too much: Hearing out neurodivergent students, universal design for learning and the case for integrating more anonymous technology in higher education. *Higher Education*. <https://doi.org/10.1007/s10734-024-01389-6>
- Durgungoz, A., & Kharrufa, A. (2025). ChatGPT is like A study buddy, A teacher and sometimes just A friend: A longitudinal exploration of students' interactions, perception and acceptance. *Interactive Learning Environments*, 1–20. <https://doi.org/10.1080/10494820.2025.2509276>
- Eichelberger, A., & Ngo, H. T. (2018). College students' perception of an online course in special education. *International Journal for Educational Media and Technology*, 12(2), 1–19.
- Göksün, D. O., & Gürsoy, G. (2019). Comparing success and engagement in gamified learning experiences via Kahoot and Quizizz. *Computers & Education*, 135, 15–29. <https://doi.org/10.1016/j.compedu.2019.02.015>
- Hernanz, V., Latorre-Coscolluela, C., Suárez, C., & Lanchares-Sancho, E. (2024). Revitalising learning in three university contexts: Unleashing the power of the Quizizz app to increase self-efficacy, intrinsic motivation, satisfaction, and performance. *Education And Information Technologies*, (17), 19. <https://doi.org/10.1007/s10639-024-12779-9>
- Hudon, A., Kiepora, B., Pelletier, M., & Phan, V. (2024). Using chatGPT in psychiatry to design script concordance tests in undergraduate medical education: Mixed methods study. *JMIR Medical Education*, 10(1), e54067. <https://doi.org/10.2196/54067>
- Hutt, S., & Hieb, G. (2024). Scaling Up Mastery Learning with Generative AI: Exploring How Generative AI Can Assist in the Generation and Evaluation of Mastery Quiz Questions. In *Proceedings of the Eleventh ACM Conference on Learning@ Scale*, 310–314. <https://doi.org/10.1145/3657604.3664699>
- Keadkraichaiwat, I., Sitticharoon, C., Letsiripatarajit, V., & Maprapho, P. (2024). Effects of using audience response systems (Kahoot and Google forms and Sheets) on learning of medical students in a large class: A questionnaire-based retrospective study. *Education and Information Technologies*, (13), 25. <https://doi.org/10.1007/s10639-024-12548-8>
- Laohawetwanit, T., Apornvirat, S., & Kantasiripitak, C. (2024). ChatGPT as a teaching tool: Preparing pathology residents for board examination with AI-generated digestive system pathology tests. *American Journal of Clinical Pathology*, , Article aqae062. <https://doi.org/10.1093/ajcp/aqae062>
- Maimaiti, G., & Hew, K. F. (2025). Gamification bolsters self-regulated learning, learning performance and reduces strategy decline in flipped classrooms: A longitudinal quasi-experiment. *Computers & Education*, 230, 105278. <https://doi.org/10.1016/j.compedu.2025.105278>
- Merrill, M. D. (2002). First principles of instruction. *Educational Technology Research and Development*, 50(3), 43–59. <https://doi.org/10.1007/BF02505024>
- Meşe, İ., Taşlıçay, C. A., Kuzan, B. N., Kuzan, T. Y., & Sivrioğlu, A. K. (2024). Educating the next generation of radiologists: A comparative report of ChatGPT and e-learning resources. *Diagnostic and Interventional Radiology*, 30(3), 163. <https://doi.org/10.4274/dir.2023.232496>
- Mushtaq, N., Nazeer, N., Fayaz, I., et al. (2025). Next-gen learning: Gamifications impact on higher education. *Education and Information Technologies*, 30, 15691–15717. <https://doi.org/10.1007/s10639-025-13431-w>
- Ngoc, H. L. T. (2024). The effectiveness of using Quizizz in teaching English as a foreign language at a tertiary level. *Journal of Knowledge Learning and Science Technology*, 3(1), 207–222. <https://doi.org/10.60087/jklst.vol3.n1.p222>
- Nicol, D. J., & Macfarlane-Dick, D. (2006). Formative assessment and self-regulated learning: A model and seven principles of good feedback practice. *Studies in Higher Education*, 31(2), 199–218. <https://doi.org/10.1080/03075070600572090>

- Park, S., & Kim, S. (2021). Is sustainable online learning possible with gamification? The effect of gamified online learning on student learning. *Sustainability*, 13(8), Article 4267. <https://doi.org/10.3390/su13084267>
- Richardson, J. C., Koehler, A. A., Besser, E. D., Caskurlu, S., Lim, J., & Mueller, C. M. (2015). Conceptualising and investigating instructor presence in online learning environments. *International Review of Research in Open and Distributed Learning*, 16(3), 256–297. <https://doi.org/10.19173/irrodl.v16i3.2123>
- Scherp, H. (2013). Quantifying qualitative data using cognitive maps. *International Journal of Research & Method in Education*, 36(1), 67–81. <https://doi.org/10.1080/1743727X.2012.696244>
- Sekli, G. M., Godo, A., & Véliz, J. C. (2024). Generative AI solutions for faculty and students: A review of literature and roadmap for future research. *Journal of Information Technology Education: Research*, 23, 014. <https://doi.org/10.28945/5304>
- Sercemeli, M., & Baydas Onlu, O. (2023). Prediction of students' learning outcomes by various variables in gamified online accounting courses. *Education and Information Technologies*, 28(12), 16979–17007. <https://doi.org/10.1007/s10639-023-11873-8>
- Stiggins, R. (1994). *Student-centered classroom assessment*. Merrill Publishing Company.
- Striepe, M. (2021). Combining concept mapping with semi-structured interviews: Adding another dimension to the research process. *International Journal of Research & Method in Education*, 44(5), 519–532. <https://doi.org/10.1080/1743727X.2020.1841746>
- Taani, O., & Alabidi, S. (2024). ChatGPT in education: Benefits and challenges of ChatGPT for mathematics and science teaching practices. *International Journal of Mathematical Education in Science and Technology*. <https://doi.org/10.1080/0020739X.2024.2357341>
- Tandiono, R. (2024). Gamifying online learning: An evaluation of kahoot's effectiveness in promoting student engagement. *Education and Information Technologies*. <https://doi.org/10.1007/s10639-024-12800-1>
- Torrance, H., & Pryor, J. (1998). *Investigative formative assessment: Teaching, learning, and assessment in the classroom*. Open University.
- Yan, L., Sha, L., Zhao, L., Li, Y., Martinez-Maldonado, R., Chen, G., Li, X., Jin, Y., & Gašević, D. (2023). Practical and ethical challenges of large language models in education: A systematic scoping review. *British Journal of Educational Technology*, 55, 90–112. <https://doi.org/10.1111/bjet.13370>
- Zainuddin, Z. (2023). Integrating ease of use and affordable gamification-based instruction into a remote learning environment. *Asia Pacific Education Review*. <https://doi.org/10.1007/s12564-023-09832-6>
- Zhang, Z., & Crawford, J. (2024). EFL learners' motivation in a gamified formative assessment: The case of Quizizz. *Education And Information Technologies*, 29, 6217–6239. <https://doi.org/10.1007/s10639-023-12034-7>
- Zhang, Z., & Huang, X. (2024). Exploring the impact of the adaptive gamified assessment on learners in blended learning. *Education and Information Technologies*, (16), 21. <https://doi.org/10.1007/s10639-024-12708-w>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Ahmet Durgungoz^{1,2}  · Fatma Canan Durgungoz^{3,4} 

✉ Ahmet Durgungoz
ahmet.durgungoz@warwick.ac.uk

¹ Computer Education and Instructional Technology, Mersin University, Mersin, Türkiye

² Computer Science, University of Warwick, Coventry, UK

³ Special Education, Mersin University, Mersin, Türkiye

⁴ Department of Psychology, University of Bath, Bath, UK